

Parakh Arena · Strategy Note

Team: IdeaForge · Harsh Kawatra · Delhi Technological University (DTU) **GitHub repository:** <https://github.com/harshkawatra11/parakh-iitd-hack> **Live review console:** <https://parakh-ideaforge.vercel.app/arena> **Full method writeup:** ARENA_README.md in this zip, or arena/ARENA_README.md in the repository above

The one rule everything follows

A credit costs `PENALTY_FACTOR` points (0.05, announced at kick-off). Every paid action, a search, a profile, a batch, an assessment, an offer, a release, only happens when `expected_points_gained > cost_in_credits x PENALTY_FACTOR`. All thresholds live in `config.json`, re-read every ten seconds, so tuning never needs a restart.

What it bought, and when

Recon and the opening wave. Per requisition: page `/search` on the role, rank summaries locally by skill overlap, fetch the best in one `/candidates/batch` call per fifty, then run every profile through the same fabrication checks that won the preliminary round (age vs. graduation, experience vs. career history). A verified `/assess` is bought before every offer, since a fabricated profile looks best on paper and a bad signing wastes a slot worth roughly ten points. By 15:18, three minutes after connecting, **all 116 slots across 10 requisitions were filled**, for 12,031 credits.

The closing-hour upgrade engine. The 15:30 snapshot showed +440 points, rank 17 of 51, 116/116 filled, but our hires sat only marginally above each bar (mean quality 0.47/1.0). The agent was extended mid-run to continuously find a stronger, independently verified replacement for its weakest held hire, release the weak one, and sign the strong one. Every swap is measured: at each 15-minute snapshot the agent computes real points gained per unit of quality improved across that window's swaps, and would switch itself off below a safety floor (8). It never did: measured slope was **36.9 at 16:00** and **34.5 at 16:15**. Result: **119 verified upgrades, zero crashes, zero failed releases**, points rising 1,042 to 2,251, all 116 slots held throughout, 49,700 of 50,000 credits spent because the return stayed far above cost the whole window.

Handling rejections

Every offer-rejection reason is handled explicitly: `already_signed` and `same_person_already_signed` retire that id, `requisition_full` stops further offers there, `role_mismatch` retires just that pairing. On an upgrade, the weak hire is released only after its replacement is already verified; if the follow-up offer is then lost to a rival, the ordinary fill pass repicks the slot within ten seconds.

Tuned live, on evidence

Two thresholds were adjusted mid-run, both config-only, no code change: the credit reserve was relaxed once the opening wave filled every slot and the closing-hour cost doubling was confirmed from measured credit deltas; and around 16:14 the upgrade engine went quiet with 12,000+ credits unspent because its own early success had pushed every requisition's weakest hire above the "worth upgrading" ceiling set for a slower improvement curve. The ceiling was raised live from the log evidence; the engine resumed within seconds and completed 49 more upgrades in the final fifteen minutes.

With two more hours

Run requisitions in parallel within the rate limit for wider recon. Read `/market` price pressure to reorder attention toward roles rivals are filling fastest. Treat the points formula as continuously recalibrated from the first snapshot rather than a single starting guess, since the true formula rewards skill and margin more steeply than our initial linear assumption, a gap the upgrade engine's measurements exposed and corrected in the second half rather than the first fifteen minutes.

Edge cases

Empty search stops that role's paging. Every exception is caught, logged with its traceback, and the loop continues; one crash occurred (a transient `/requisitions` network exhaustion), self-healed after a five-second backoff. Exhausted budget is a soft stop: the agent idles on free `/ledger` polls rather than erroring. Rate limiting is handled inside the provided `arena_client.py`, used unmodified. State is checkpointed after every signing and swap, so the one deliberate redeploy (to add the upgrade engine) resumed exactly where it left off with no re-spending and no duplicate offers.

AI assistants declared

Anthropic Claude (Claude Code) was used to draft and refactor the agent's code and this note, disclosed the same way round one disclosed it.

Final ledger: 116/116 slots filled · 119 verified upgrades · 1 self-healed crash, zero fatal · 2,251 points at the last snapshot (16:30) · 49,700 of 50,000 credits used · score +266 at 16:00, rising through further verified upgrades to the close.